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Outline

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Highlights

- An isopropanol - polyvinylpyrrolidone - ammonia - sodium citrate system for preparation of SiO₂ wires.
- The present system can significantly reduce both the amount of PVP and the preparation time.

Abstract

This study presents a method for the production of SiO₂ wires in a system composed of isopropanol, polyvinylpyrrolidone, ammonia and sodium citrate. The typical wires were mildly curved with an average diameter of 500 nm and an aspect ratio up to 60 or more.

A 50-fold expansion of this process still gave stable wires with a yield of about 86 %. Monitoring the sodium levels along the SiO₂ wires revealed that sodium was present in different parts of the wires. Based on this phenomenon, we propose a growth mechanism for SiO₂ wires. The present system produces SiO₂ wires more stably and efficiently than the previously reported n-pentanol-PVP-ammonia-sodium citrate method, while significantly reducing both the amount of PVP and the preparation time.

Keywords

SiO₂ wires; Wet chemical method; Emulsion; Sodium citrate; Polyvinylpyrrolidone

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1. Introduction

SiO₂ wire or rod is an important inorganic non-metallic material known for its low density, high aspect ratio [1,2], and unique microstructural effects such as surface effect, quantum tunneling, and small size effect conferred by its microstructure [3,4]. These properties make it applicable in pharmaceuticals [5,6], electronic batteries [7], biomaterials, and other applications [[8], [9], [10], [11]].

Great efforts have been made to explore new techniques to develop SiO₂ wires or rods, including chemical vapor deposition [[12], [13], [14]], electrostatic spinning [15,16], the hydrothermal method [17,18], thermal oxidation [19], metal catalysis [20,21], electrochemical process [22], and wet chemical process using various chemicals as a template, such as cylindrical polymer brushes(CPBs) [23,24], emulsion [[25], [26], [27]], gold nanoparticle [28], and peptides [29].

[Declaration of competing interest](#)

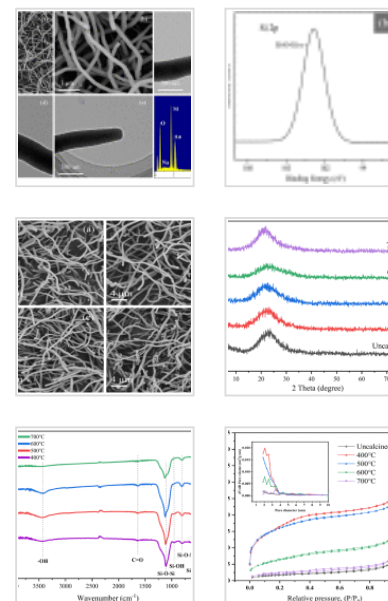
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Preparation of SiO₂ wires or rods by wet chemical process remains attractive and challenging, but several studies have made progress. Kuijk [30] et al. developed a PVP-sodium citrate-pentanol system for synthesis of monodisperse rodlike silica colloids with diameters of 200 nm and greater and lengths up to 10 μm, resulting in aspect ratios from 1 to 25. The growth mechanism of these rods involves emulsion droplets inside which silica condensation takes place. Due to an anisotropic supply of reactants, the nucleus grows to one side only, resulting in rod formation. By using such a PVP-sodium citrate-pentanol system, Jaswinder Sharma [31] et al. demonstrated the manipulation of the reaction temperature to achieve control of the local thickness of already growing silica rods. Furtherly, Yi [32] et al. prepared a mechanically robust and flexible Alkyl-capped silica nanowires, with a diameter of approximately 80 nm and tunable aspect ratios (1–150) using a PVP-sodium citrate-pentanol system by introducing the surfactant-like hydrolysate of tri-methoxy(octadecyl)silane (C₁₈TMOS). Due to the interfacially arranged C₁₈Si(O)₃ reduces the surface tension of the water droplet, the initial water droplet is deformed into a more elongated cone-like structure and then forms a tadpole-like silica nanorod as particle growth progresses. With the continuous hydrolysis/condensation reaction of TEOS, silica nanowires are eventually developed, and grows to its full length when the added TEOS is consumed.

Although the water-in-alcohol emulsion system has shown incomparable versatility in the controlled synthesis of anisotropic silica micro-/nanostructures, the exact roles of the alcohol phase, PVP and sodium citrate in the anisotropic growth of silica particles still remain elusive. Yu [33] et al. proposed a “salting-out” mechanism to explain the formation of a Janus-like silica-droplet structure at the initial silica condensation stage, which is essential for the present anisotropic growth mode.

Inspired by the results of the above research work, we present here an isopropanol -PVP-ammonia -sodium citrate system for the synthesis of submicron SiO₂ wires with a large aspect ratio up to 60 or more. In this system, SiO₂ wires can be stably produced within 2 h with a high yield up to 90 %. A 50-fold expansion of this process still gave stable wires with a yield of about 86 %. In this system, some phenomena are different from those in

the PVP-sodium citrate-pentanol system. We believe that these results would lead to new insights into the anisotropic growth of silica particles.

2. Experimental detail

2.1. Materials

Polyvinylpyrrolidone (PVP, Mw=58,000, Shanghai Macklin Biochemical Technology Co., Ltd). Sodium citrate (Shanghai Aladdin Chemistry Co., Ltd). Tetraethyl orthosilicate (TEOS), isopropanol (IPA) and ammonia (wt %=25 %–28 %, Chengdu Kelong Chemical Co., Ltd) were all used as received without any purification. Deionized water with a resistivity of 18.0 MΩ was used in all the experiments.

2.2. Preparation of SiO₂ wires

By systematically optimizing key parameters—including reaction temperature (30 °C–80 °C), sodium citrate dosage (50 µL–550 µL), ammonia dosage (100 µL–450 µL) and PVP dosage (0 g–0.3 g), an optimized and typical process was developed. The typical process is as follows:

0.03 g of PVP was introduced into a 10 mL polyethylene tube containing 5 mL of isopropanol and the mixture was sonicated until complete dispersion of the PVP was achieved. Then 250 µL of deionized water, 300 µL of sodium citrate (0.18 M), and 200 µL of ammonia (10 M) were added successively to the tube and manually shaken for 2 min. 300 µL of TEOS was then added to the tube and shaken well. Finally, the tube was placed in a 75 °C water bath and heated for 2 h.

The tubes were taken out of the water bath, allowed to cool to room temperature, and then centrifuged to separate the solids from the liquid. The solid obtained was washed five times each with water and ethanol, and then dried at 80 °C for 5 h.

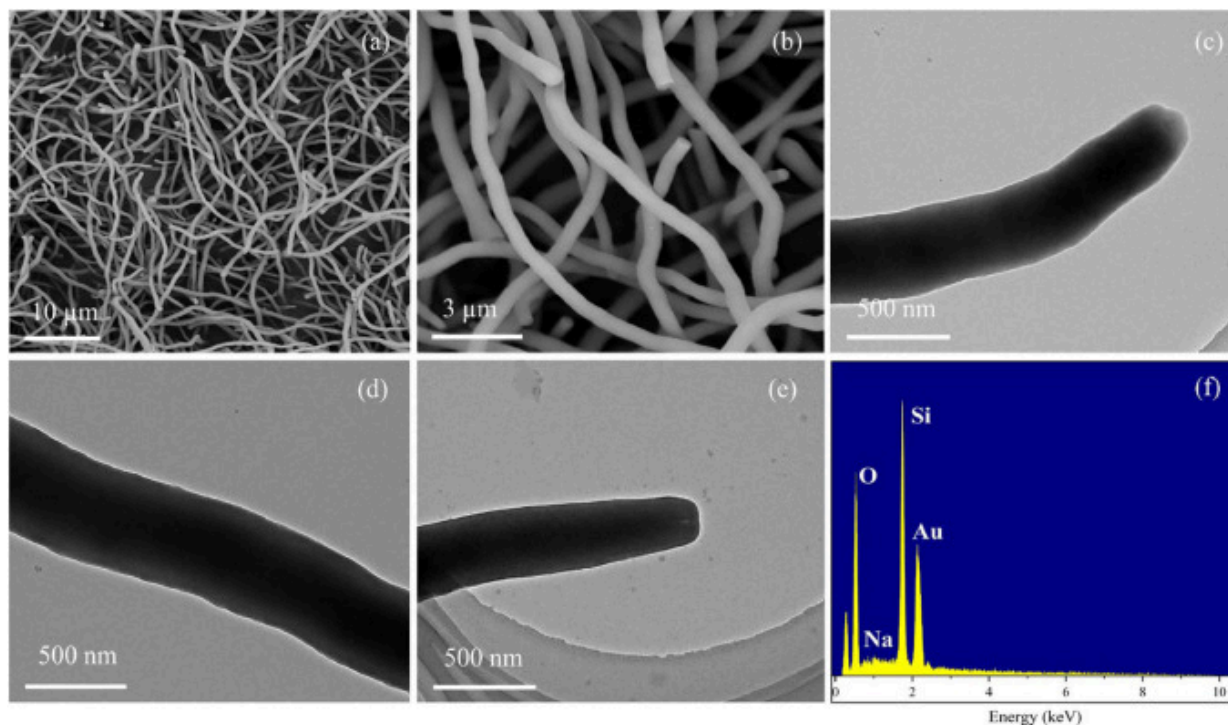
2.3. Characterization

Field emission scanning electron microscope (SEM, FEI Quattro S, USA) and transmission electron microscope (TEM, JEOL Jem-f200, Japan) were used to observe the morphology of the synthesized samples; the specific surface area and pore size distribution of the materials were analyzed and determined by using a specific surface area and pore size analyzer (BSD-PM1/2, Best Instruments, Inc., China); the surface chemical functional groups of the material were determined using a fourier transform infrared spectrometer (FTIR, IS5Nicolet, USA); the distribution of elements and Na content was analyzed using an energy dispersive spectrometer (EDS, BRUKER XFlash 6130, German); the crystalline structure of the materials was analyzed using X-ray diffraction(XRD, DX-2700, China); the chemical compositions were analyzed using X-ray photoelectron spectroscopy (XPS, Thermo Fisher Scientific-K-Alpha, USA).

3. Results and discussion

3.1. Characteristics of SiO₂ wires

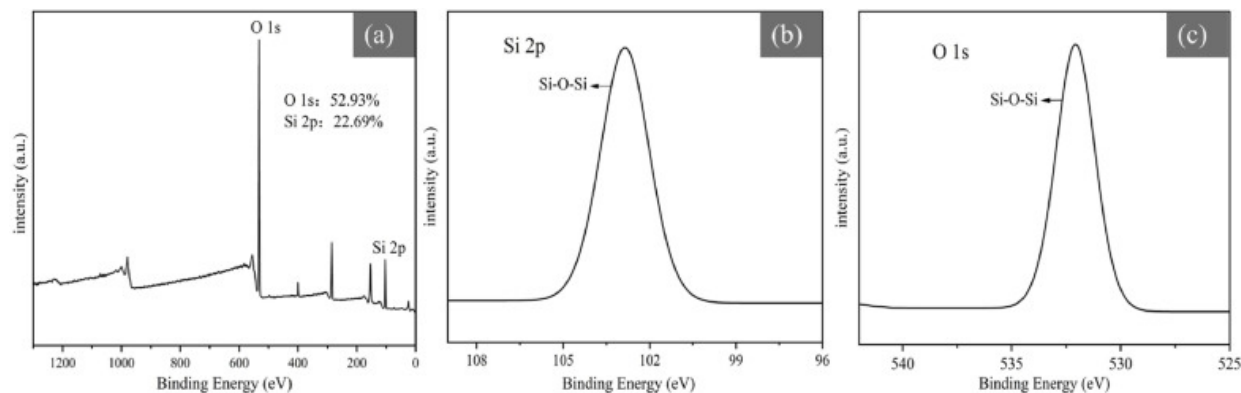
[Fig. 1](#) shows the appearance and elemental analysis result of a product prepared with the optimized process. The typical wires were mildly curved long wires with an average diameter of 500 nm (Standard deviation=26.5 nm, [Fig. 1a](#)) and a length of a few tens of micrometers, and an aspect ratio up to 60 or more (aspect ratio = length/diameter). The entire wire is solid, and the diameter of the tail is smaller than that of the middle ([Fig. 1b-e](#)). The EDS analysis result indicates that the wires mainly consist of Si and O ([Fig. 1f](#)) and further XPS analysis showed that the atomic percentage of Si was 22.69 %, the atomic percentage of O was 52.93 % ([Fig. 2a](#)). The XPS Si 2p nuclear level positioned at around 102.9 eV corresponds to elemental Si in SiO₂, indicating that Si is present in the material in the form of Si⁴⁺, and the peak at 532.1 eV in the O1 s core level is also attributed to SiO₂ [[17,34](#)]([Fig. 2b and c](#)).



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Fig. 1. Morphology and elemental analysis result of SiO₂ wires. (a, b) SEM images of SiO₂ wires; (c-e) Transmission electron microscopy images of the head, middle, and tail of a single wire, respectively; (f) EDS plot of material tested after gold spraying. Experimental conditions: 0.03 g PVP, 5 ml isopropanol, 250 μ L deionized water, 300 μ L sodium citrate (0.18 M), 200 μ L ammonia (10 M), 300 μ L TEOS, and 75 $^{\circ}$ C for 2 h.

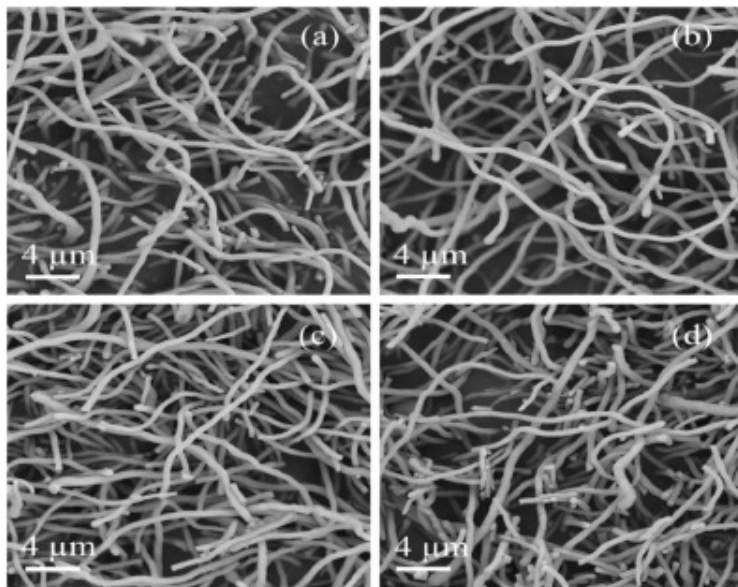


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Fig. 2. XPS spectra of (a) survey spectra, high resolution spectrum of Si 2p (b) and O 1s (c).

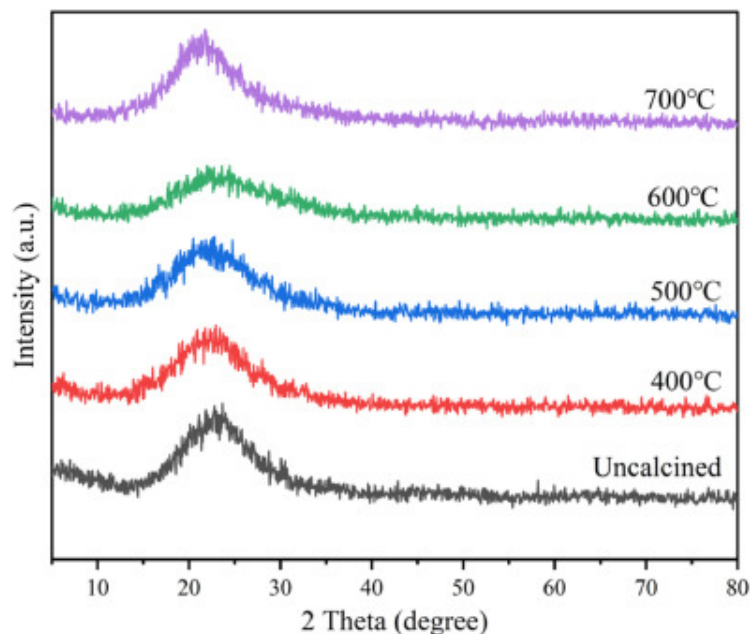
In order to remove the residual PVP and sodium citrate from the SiO₂ wires, the prepared SiO₂ wires were calcined at high temperatures from 400 °C to 700 °C. The SiO₂ wires calcined at 400 °C or 500 °C for 1 h still maintained their morphology well, however, the diameter became smaller (about 400 nm, Fig. 3a and b), while the wires calcined at 600 °C or 700 °C for 1h partially fragmented due to its further shrinkage (Fig. 3c and d). The spectra in Fig. 4 show a strong and broad diffraction peak between 15 and 30° for both calcined and uncalcined material, indicating that the SiO₂ wires are amorphous [35,36].



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Fig. 3. SEM images of SiO₂ wires calcined at 400 °C (a), 500 °C (b), 600 °C (c), 700 °C (d).

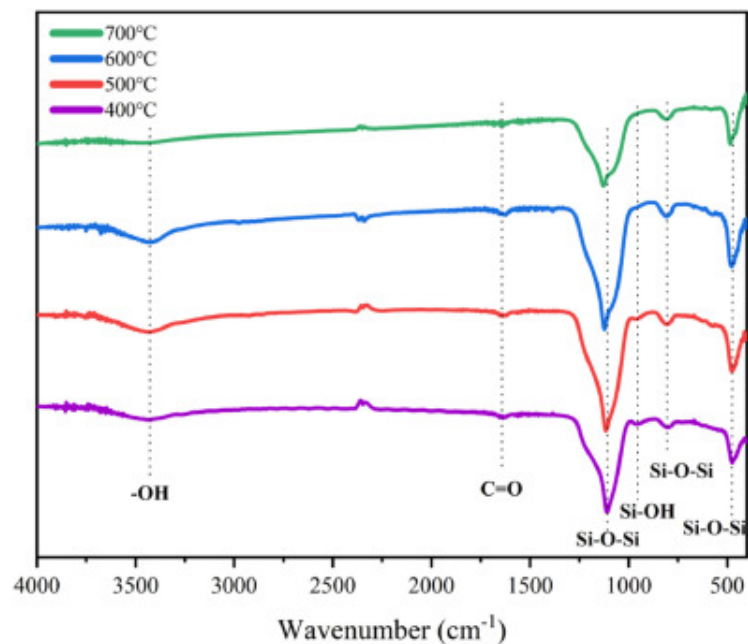


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Fig. 4. XRD images of SiO₂ wires calcined at 400 °C, 500 °C, 600 °C, 700 °C and uncalcined.

Fig. 5 illustrates the infrared spectra of SiO₂ wires calcinated at different calcination temperatures. The peak at 3436.5 cm⁻¹ shows stretching vibrations of the -OH bond; peak at 1645.0 cm⁻¹ shows stretching vibrations of the C = O bond, which is the characteristic peak of the PVP [37]; peak at 1105.0 cm⁻¹, 804.2 cm⁻¹ and 472.5 cm⁻¹ shows the antisymmetric stretching vibration, bending vibration and wobbling vibration of Si-O-Si; peak at 962.3 cm⁻¹ shows the antisymmetric stretching vibration of Si-OH [38,39]. With the increase of calcination temperature, the characteristic peak of PVP disappeared at 700 °C, which proved that the residual PVP had been removed and pure SiO₂ wires were obtained.

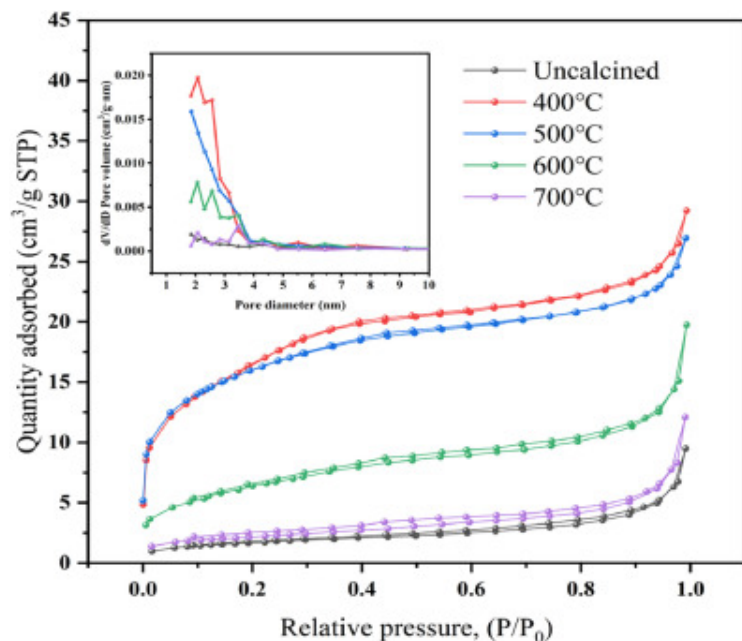


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Fig. 5. Infrared spectra of SiO₂ wires calcined at 400 °C, 500 °C, 600 °C and 700 °C.

The Brunauer-Emmett-Teller (BET) method was used to determine the specific surface area of SiO₂ wires, and the Barrett-Joyner-Halenda (BJH) model was used to calculate the pore size distribution (Fig. 6, for detailed data see S1 in the supporting information).



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Fig. 6. N₂ adsorption and desorption curves and pore size distribution of SiO₂ wires calcined at 400 °C, 500 °C, 600 °C, 700 °C and uncalcined.

The specific surface area of the wires uncalcined and the wires calcined at 400 °C is 5.91 m²/g and 57 m²/g, respectively. This is mainly due to the removal of PVP and water from the wires, increasing holes in the wires. As the calcination temperature increases from 400 °C to 700 °C, the specific surface area decreases from 57 m²/g to 8 m²/g. This may be due to the shrinkage of wires resulting from further condensation of the residual Si-OH.

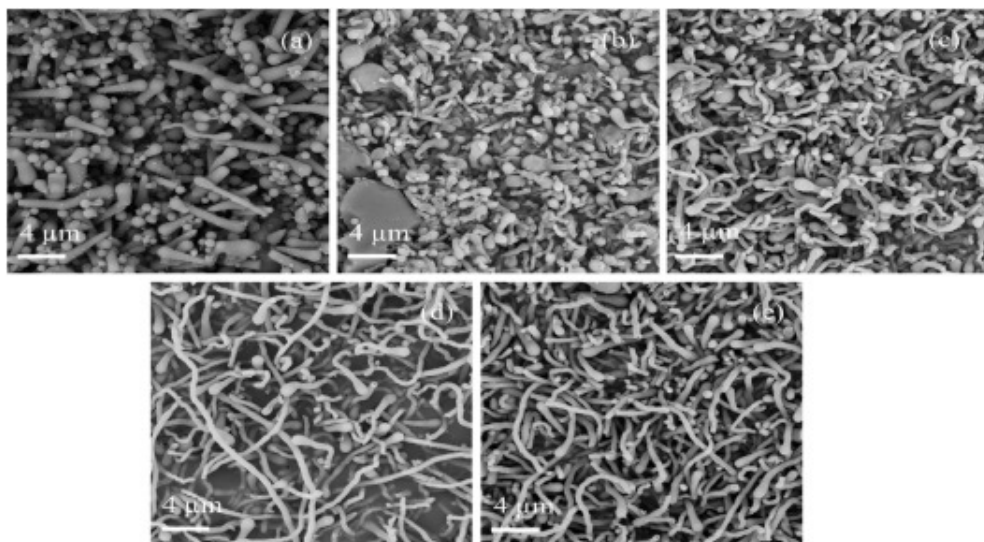
All isotherms show a small hysteresis between adsorption and desorption, and type IV isotherm hysteresis curve [40]. Compared to these SiO₂ wires calcined at 600 °C and 700 °C, the SiO₂ wires calcined at 400 °C and 500 °C show an increasing trend in the adsorption curves at lower relative pressures (P/P₀ < 0.01). Although the curves here show the presence of meso- and microporous structures in the material, the small specific

surface area and the transmission electron microscopy picture (Fig. 1 c-e) suggest that these pores are rare.

3.2. Changes in morphology with preparation conditions

In this work, the effects of synthesis conditions on the morphology and size of SiO₂ wires were sufficiently investigated.

Fig. 7 shows the morphology of SiO₂ wires prepared at temperatures from 30 °C to 80 °C (S2 in the supporting information). At a temperature of 30 °C, spherical particles and rods with spherical heads were obtained; tadpole-shaped, curved and hollow products were obtained at temperatures of 50 °C and 60 °C; at temperatures of 75 °C and 80 °C, SiO₂ wires with spherical heads were obtained.

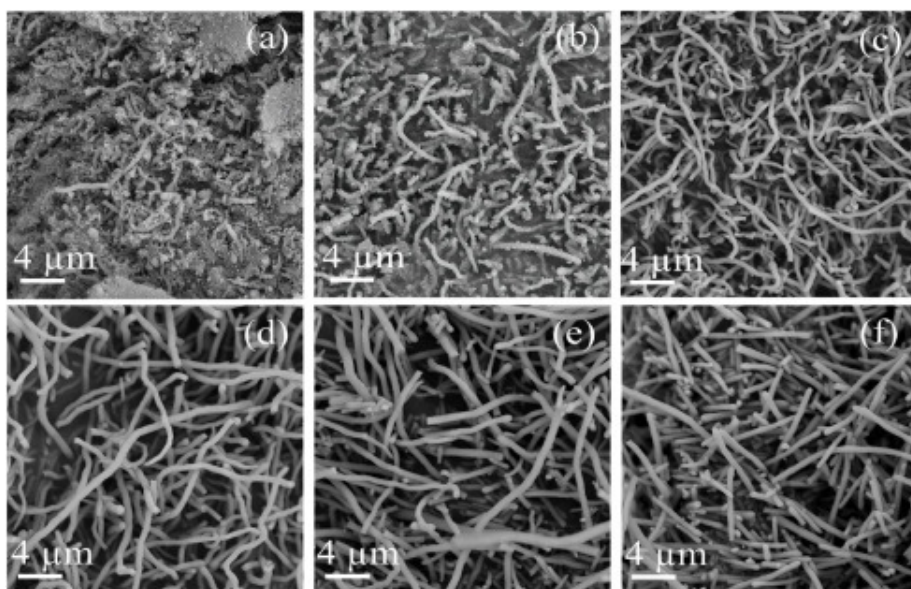


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Fig. 7. Morphology of SiO₂ wires obtained at temperatures of 30 °C (a), 50 °C (b), 60 °C (c), 75 °C (d) and 80 °C (e).

Fig. 8 illustrates the effects of sodium citrate dosage on the morphology of the product. When the amount of sodium citrate was 50 μL and 100 μL , a granular product and a small amount of wires were obtained. This may be due to the insufficient amount of sodium citrate and thus most TEOS hydrolysis and condensation occur within the isopropanol phase. However, almost entirely wires could be obtained when the amount of sodium citrate exceeded 200 μL . In this case, due to the presence of nuclei derived from sodium citrate in solution, most TEOS could be deposited onto the existing nuclei, thus facilitating wire growth. When the amount of sodium citrate exceeded 500 μL , short, straight rods with smaller diameters were obtained (S3 in the supporting information).



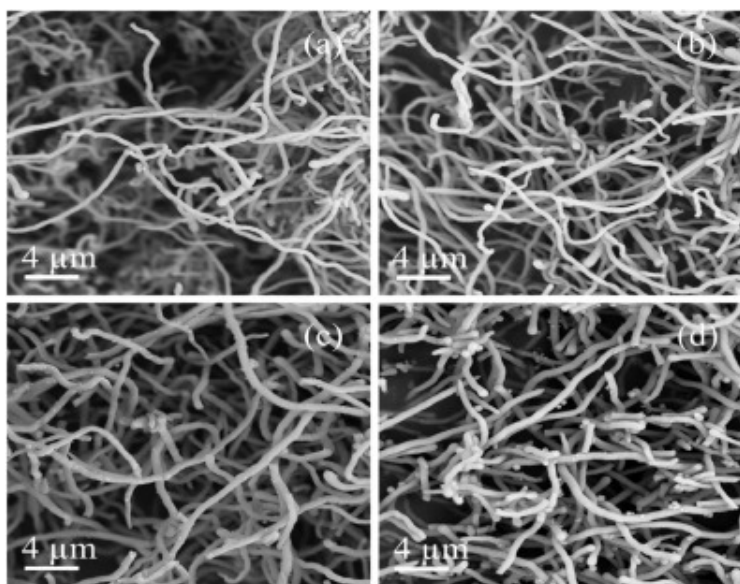
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Fig. 8. Morphology of product prepared with 50 μL (a), 100 μL (b), 200 μL (c), 300 μL (d), 400 μL (e) and 550 μL (f) of sodium citrate.

As a catalyst, ammonia affects the rate of hydrolysis and condensation of TEOS. As can be seen in Fig. 9a and b, the lower dose of ammonia resulted in curved and longer wires with

a smaller diameter (S4a and b). Conversely, more ammonia led to shorter wires with larger diameters (Fig. 9c and d) (S4c and d).



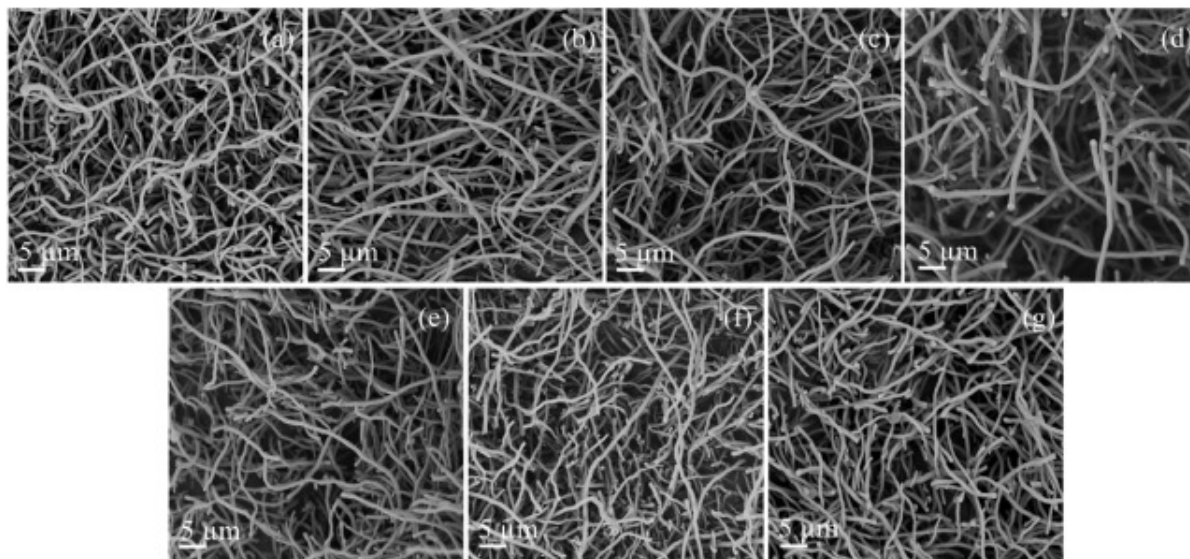
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Fig. 9. Morphology of SiO₂ wires prepared with 100 μ L (a), 200 μ L (b), 300 μ L (c), 450 μ L of ammonia.

The present system allows for the efficient preparation of SiO₂ wires, and notably, the amount of PVP utilized is reduced to just one-seventeenth and the time required was decreased to one-tenth, compared to that of the previous method [26,30,41].

In order to evaluate the stability of SiO₂ wires under various conditions, the wires were immersed in methanol, ethanol, hexane, 0.1 M hydrochloric acid, 0.1 M sodium hydroxide and distilled water for 12h. Fig. 10 shows the overall morphology of the SiO₂ wires is intact, except for a slight fracture in hexane, which proves that the SiO₂ wires are highly stable in the selected solvents (S5 in the supporting information).



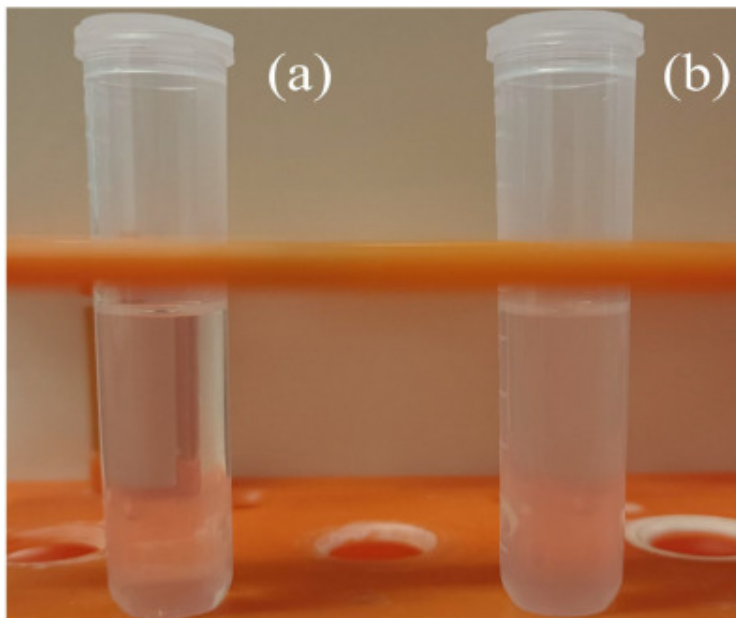
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Fig. 10. Morphology of SiO₂ wires after immersion in selected solvents for 12 h. Untreated (a); Methanol (b); Ethanol (c); Hexane (d); 0.1 M hydrochloric acid (e); 0.1 M sodium hydroxide (f); Distilled water (g).

3.3. Growth mechanism of SiO₂ wires

In the present system, introducing a small quantity of water or PVP into isopropanol does not lead to the formation of an emulsion due to the hydrophilic nature of isopropanol. In contrast, when sodium citrate is added to isopropanol, turbid emulsions are created as a result of sodium citrate's limited solubility in isopropanol (Fig. 11). This is not the same as the previous synthesis of silica nanowires in n-amyl alcohol, in which the emulsion was formed by combining a high concentration of PVP in water with sodium citrate [30,42,43]. In the present isopropanol system, when sodium citrate is omitted, the outcome is the formation of silica microspheres rather than SiO₂ wires. This underscores the crucial role of sodium citrate in facilitating the anisotropic growth of silica wires.

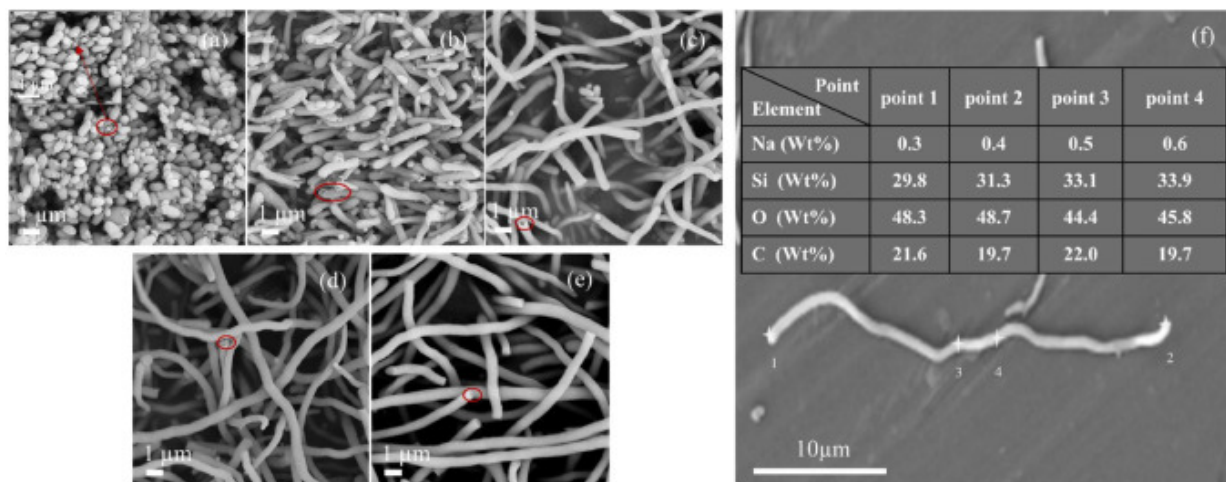


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Fig. 11. Mixture of isopropanol-PVP-TEOS- ammonia without sodium citrate (a) and with sodium citrate (b).

We photographed and documented the growth of the SiO₂ wires (Fig. 12a-e). Similar to what has been reported in the literature [32,41], the process went through an initial hollow ellipsoid, to a hollow wire, and finally to a solid wire. The formation of these wires is thought to be due to the formation of anisotropic emulsion droplets and the subsequent one-sided supply of new silicon sources.



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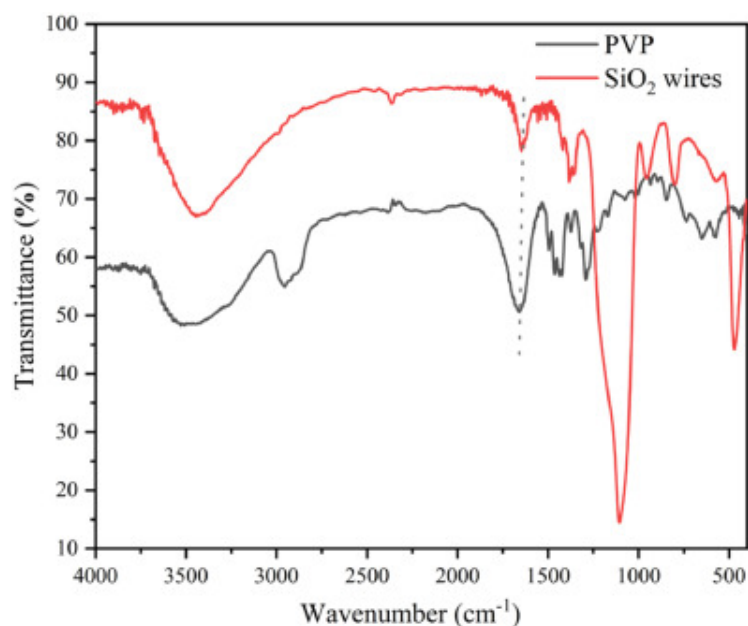
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Fig. 12. The growth process and the distribution of sodium on SiO₂ wires. (a) 5 min, (b) 10 min, (c) 30 min, (d) 60 min, (e) 120 min, (f) EDS spot-scanning of Na, Si, O, and C in different parts of a single wire.

Given the significance of sodium citrate in the wire formation process, we measured the sodium levels in various parts of the wire (Fig. 12f). As expected, sodium was found in all the measured sections. This implies that sodium citrate may shift as the wire expands instead of remaining in its original position. This is understandable because the emulsion contains water within which some of the sodium citrate will be dissolved, and as the wire grows longer, the water within the emulsion along with the sodium citrate dissolved therein will be expelled to one end, thus dispersing to each part of the wire as it grows. In theory, if the droplet is always at one end of the wire and all other necessary conditions are met, the wire can continue to grow [30].

The peak at 1645 cm⁻¹ in the infrared spectrum shows the presence of PVP in the SiO₂ wires (Fig. 13). Previous reports have suggested that polyvinylpyrrolidone has the ability to stabilize emulsions and promote the growth of silica nanowires [44,45]. In the current

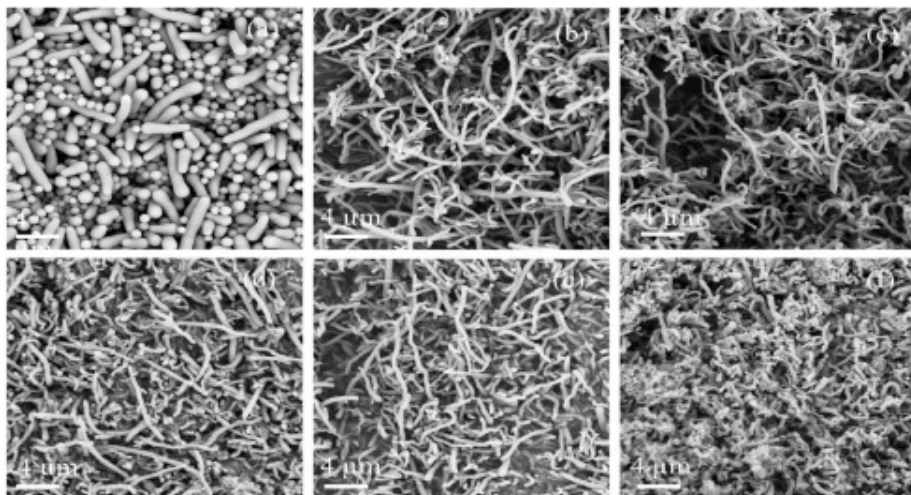
system, without the addition of PVP, rods with rounded surfaces and larger diameters (about 1 μm) are obtained, as well as spherical particles of varying sizes (Fig. 14). In contrast, smaller diameter SiO₂ wires can be produced by adding a small amount of PVP (S6 in the supporting information). This shows that PVP can also control the lateral growth of SiO₂ wires. This may be due to the fact that the PVP attached to the wire surface prevents further deposition of TEOS from the side of the wires.



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Fig. 13. Infrared spectra of PVP and prepared SiO₂ wires.



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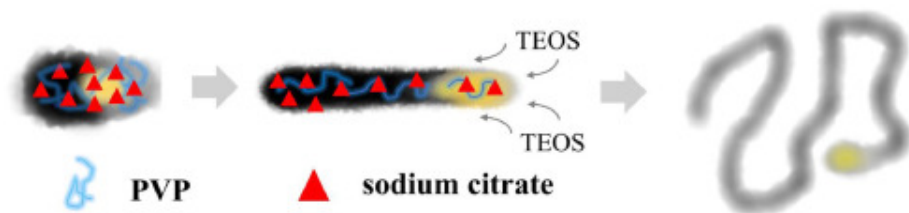
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Fig. 14. The effect of PVP amount on the morphology of SiO₂ wires. (a) 0g , (b) 0.03g , (c) 0.05g , (d) 0.07g , (e) 0.1g , (f) 0.3g.

The wire growth process can be divided into three main steps: TEOS hydrolysis on the emulsion droplet surface, transfer of hydrolysis products to the emulsion droplet's interior, and subsequent condensation and deposition within the emulsion droplet. The transfer of hydrolysis products from the droplet interface to the inside of the droplet is crucial. If this transfer rate is significantly slower than the deposition and condensation rate, TEOS will predominantly deposit on the emulsion droplet's surface. Given PVP's excellent phase-transfer ability, it's plausible that PVP at the droplet interface acts as a phase-transfer reagent, aiding the transport of hydrolysis products inward [46,47].

Taking the above conclusions together, we propose here a growth mechanism for SiO₂ wires (Fig. 15). Firstly, due to the limited solubility of sodium citrate in isopropyl alcohol, the addition of sodium citrate creates emulsion droplets that are anisotropic, and contain more water at its one end. This anisotropy is crucial for subsequent wire growth. Upon addition of TEOS, primary hydrolysis of TEOS occurs at the droplet interface where

ammonia is present. Next, the hydrolysis products were rapidly transferred to the interior of the emulsion droplets with the help of PVP. As the primary hydrolysis products migrate inside the droplet and further condense, the space inside the droplet is gradually occupied by the solid phase so that the droplet is naturally pushed to one end of the wires. Therefore, a slightly larger, nearly spherical head of a slightly larger diameter is always present in the head of the wire as it grows. If the supply of TEOS is sufficient and all other necessary conditions are present, the wires will continue to grow and eventually longer wires can be obtained.

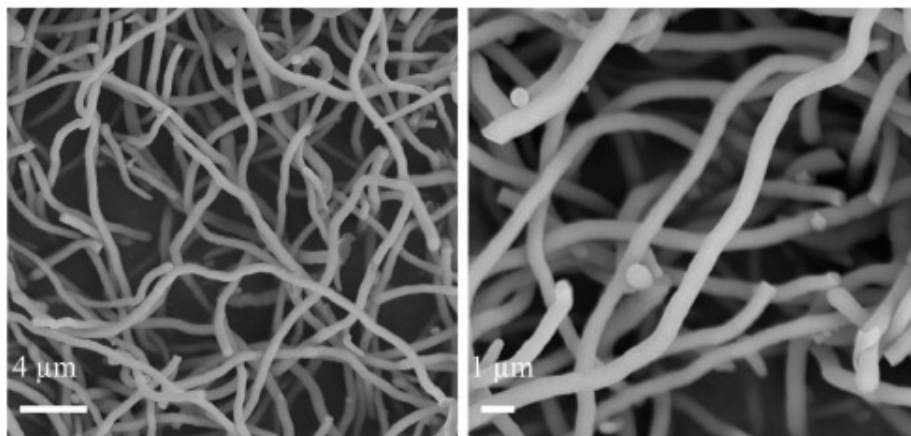


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Fig. 15. Possible growth mechanisms of SiO₂ wires.

Under optimized conditions, a 50-fold expansion of this process, with other reagents in the same proportions (S7 in the supporting information), still gave stable wires with a yield of about 86 % (Fig. 16) (S8 in the supporting information).



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Fig. 16. SiO₂ wires obtained from the scaled-up experiment.

The above results suggest that the present process could enable the large-scale production of SiO₂ wires. The stability of the resulting SiO₂ wires indicates its potential applications in battery and membrane manufacturing, as well as in separation and extraction field, etc.

4. Conclusion

In summary, SiO₂ wires with an average diameter of 500 nm and aspect ratio up to 60 or more were synthesized in an isopropanol-polyvinylpyrrolidone-sodium citrate system with a yield of about 90 %. A 50-fold scaling up of this process still produced wires with a yield of about 86 %. XRD results indicated the product was amorphous SiO₂. The as-prepared SiO₂ wires showed good stability in different solvents, and at higher temperature up to 700 °C. Results showed as the wires grow, sodium citrate is dispersed throughout the wires. Based on this, we tend to believe that the movement of sodium citrate was relative with the wire growth. The present system could produce SiO₂ wires

stably and efficiently, while significantly reducing both the amount of PVP and the preparation time compared to that of the previous method.

Data availability

Data will be made available on request.

CRediT authorship contribution statement

Tianfeng Yang: Writing – original draft. **Dongxue Ren:** Data curation. **Xiaping Zhu:** Data curation. **Qing Li:** Data curation. **Haiyan Li:** Data curation. **Jia Chen:** Data curation. **Hongyi Sun:** Data curation. **Yonghua Sun:** Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix. Supplementary materials

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